GRASSMANNIANS AS MODULI SPACES: REPRESENTABILITY, UNIVERSAL EXACT SEQUENCES, AND PROJECTIVE GEOMETRY

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Abstract. We study Grassmannians over an arbitrary base scheme from the functor-of-points point of view. Let S be a scheme, let E be a finite locally free $\mathcal{O}_S$-module of rank n, and fix an integer d with $0 < d < n$. We consider the moduli functor that sends an S-scheme T to the set of locally free quotients $E_T \twoheadrightarrow Q$ of rank $n-d$, or equivalently to rank d subbundles of E_T with locally free quotient. We prove that this functor is representable by a scheme $\mathrm{Gr}_S(d, E)$ by constructing standard open subfunctors represented by affine spaces and gluing them along explicit transition maps. This construction produces the universal exact sequence on $\mathrm{Gr}_S(d, E)$ and yields its classifying property and compatibility with base change. We then describe the relative tangent bundle in terms of the universal subbundle and quotient bundle, and over a field we recover the classical Plücker morphism and the basic projective geometry of the Grassmannian.

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1. Functors of points and representability in the background

This section records the basic language needed for the moduli-theoretic construction of the Grassmannian. We keep the discussion brief and restrict attention to the notions that will be used later.

1.1. Categories, contravariant functors, and natural transformations.

We begin with the minimal categorical framework.

Definition 1.1. A *category* $\mathcal{C}$ consists of a class of objects, a set $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ of morphisms for each pair of objects X, Y, identity morphisms $\mathrm{id}_X \in \mathrm{Hom}_{\mathcal{C}}(X,X)$, and composition laws

$$\mathrm{Hom}_{\mathcal{C}}(Y,Z) \times \mathrm{Hom}_{\mathcal{C}}(X,Y) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X,Z), \qquad (g,f) \longmapsto g \circ f,$$

satisfying the usual associativity and identity axioms.

Definition 1.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories.

(1) A *covariant functor* $F : \mathcal{C} \to \mathcal{D}$ assigns to each object X of $\mathcal{C}$ an object $F(X)$ of $\mathcal{D}$, and to each morphism $f : X \to Y$ a morphism $F(f) : F(X) \to F(Y)$, in a way compatible with identities and composition.
(2) A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$, where $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$ and all arrows reversed.

Definition 1.3. Let $F, G : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}$ be contravariant functors. A *natural transformation* $\eta : F \Rightarrow G$ consists of morphisms

$$\eta_X : F(X) \longrightarrow G(X)$$

for all objects X of $\mathcal{C}$ such that for every morphism $f : Y \to X$ in $\mathcal{C}$ the square

$$\begin{tikzcd} F(X) \arrow[r, "\eta_X"] \arrow[d, "F(f)"'] & G(X) \arrow[d, "G(f)"] \\ F(Y) \arrow[r, "\eta_Y"'] & G(Y) \end{tikzcd}$$

commutes.

Remark 1.4. In this article the main objects of study are set-valued contravariant functors on categories of schemes. One should think of such a functor F as assigning to each test scheme T a set $F(T)$ of geometric data parametrised by T, together with pullback maps along morphisms $T' \to T$.

1.2. **Representable functors and Yoneda.** Representable functors are the bridge between schemes and moduli problems.

Definition 1.5. Let $\mathcal{C}$ be a category and let X be an object of $\mathcal{C}$. The *functor of points* of X is the contravariant functor

$$h_X : \mathcal{C}^{\mathrm{op}} \longrightarrow \mathbf{Set}, \qquad h_X(T) = \mathrm{Hom}_{\mathcal{C}}(T, X),$$

whose action on morphisms is given by precomposition. A contravariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is called *representable* if there exist an object X of $\mathcal{C}$ and a natural isomorphism

$$h_X \xrightarrow{\sim} F.$$

In this situation we say that X *represents* F.

The basic result is Yoneda's lemma.

Theorem 1.6 (Yoneda lemma)**.** *Let $\mathcal{C}$ be a locally small category, let $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ be a contravariant functor, and let X be an object of $\mathcal{C}$. Then there is a natural bijection*

$$\mathrm{Nat}(h_X, F) \xrightarrow{\sim} F(X),$$

sending a natural transformation $\eta : h_X \Rightarrow F$ to the element $\eta_X(\mathrm{id}_X) \in F(X)$.

Proof. Given a natural transformation $\eta : h_X \Rightarrow F$, define

$$\Phi(\eta) := \eta_X(\mathrm{id}_X) \in F(X).$$

Conversely, if $\xi \in F(X)$, define a natural transformation $\eta^\xi : h_X \Rightarrow F$ by

$$\eta^\xi_T : \mathrm{Hom}_{\mathcal{C}}(T, X) \longrightarrow F(T), \qquad f \longmapsto F(f)(\xi).$$

This is natural in T because for any morphism $g : T' \to T$ we have

$$F(g)\big(\eta^\xi_T(f)\big) = F(g)\big(F(f)(\xi)\big) = F(f \circ g)(\xi) = \eta^\xi_{T'}(f \circ g).$$

It is immediate that the constructions $\eta \mapsto \Phi(\eta)$ and $\xi \mapsto \eta^\xi$ are inverse to each other. □

A first consequence is that representability determines the representing object uniquely.

Corollary 1.7. *If a contravariant functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ is representable in two ways, say*

$$F \cong h_X \qquad \text{and} \qquad F \cong h_Y,$$

then X and Y are uniquely isomorphic.

Proof. An isomorphism $h_X \cong h_Y$ is in particular a natural transformation $h_X \Rightarrow h_Y$. By Theorem 1.6, such natural transformations correspond bijectively to morphisms $X \to Y$. Applying the same argument in the opposite direction gives a morphism $Y \to X$, and the two are inverse because the corresponding natural transformations are inverse. □

Remark 1.8. This is the basic formal reason that moduli problems are encoded by functors. One first defines the functor that records the families one wants to classify, and then asks whether that functor is representable. If it is, the representing scheme is the desired moduli space.

1.3. **Functors on** (Sch/S)**.** We now specialise to the setting relevant for algebraic geometry.

Notation 1.9. Let S be a scheme.

(1) We write Sch for the category of schemes.
(2) We write (Sch/S) for the category of schemes over S. Its objects are morphisms $T \to S$, and its morphisms are morphisms of schemes over S.

Definition 1.10. If $X \to S$ is an S-scheme, its *functor of points over* S is the contravariant functor

$$h_X : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}, \qquad h_X(T) = \mathrm{Hom}_S(T, X).$$

Remark 1.11. The value $h_X(T)$ is the set of T-valued points of X. More generally, if

$$F : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is any contravariant functor, then an element of $F(T)$ should be thought of as a family of geometric objects parametrised by T. Passing from T to another S-scheme T' by a morphism $f : T' \to T$ produces the pullback family $f^*(\xi) \in F(T')$. This is the language in which moduli problems are most naturally formulated.

1.4. **Two motivating examples.** We conclude with two basic examples that illustrate how familiar scheme constructions can be recovered from their functors of points.

Example 1.12 (Affine space)**.** Let $S = \mathrm{Spec}(R)$ and let

$$\mathbb{A}^n_S = \mathrm{Spec}(R[x_1, \ldots, x_n]).$$

For every S-scheme T there is a natural bijection

$$\mathrm{Hom}_S(T, \mathbb{A}^n_S) \cong \Gamma(T, \mathcal{O}_T)^{\oplus n}.$$

Thus a T-valued point of affine n-space is the same thing as an n-tuple of global functions on T.

Proof. By the universal property of the relative affine scheme $\mathrm{Spec}(R[x_1, \ldots, x_n])$, a morphism $T \to \mathbb{A}^n_S$ over S is equivalent to a morphism of $\mathcal{O}_S$-algebras

$$R[x_1, \ldots, x_n] \longrightarrow \Gamma(T, \mathcal{O}_T)$$

extending the structure map $R \to \Gamma(T, \mathcal{O}_T)$. Such a morphism is determined uniquely by the images of the coordinates $x_1, \ldots, x_n$. □

Example 1.13 (Fiber products)**.** Let $X \to S$ and $Y \to S$ be S-schemes. Then for every S-scheme T there is a natural bijection

$$\mathrm{Hom}_S(T, X \times_S Y) \cong \mathrm{Hom}_S(T, X) \times \mathrm{Hom}_S(T, Y).$$

In other words, the functor of points of the fiber product $X \times_S Y$ is the product of the functors of points of X and Y.

Proof. This is precisely the universal property of the fiber product. A map $T \to X \times_S Y$ over S is equivalent to a pair of maps $T \to X$ and $T \to Y$ over S, and this correspondence is functorial in T. □

Remark 1.14. These two examples already show the pattern we will use throughout the paper. One describes a geometric object by specifying its functor of points, and one identifies a moduli space by proving that the relevant moduli functor is representable.

2. Sheaf conditions, open subfunctors, and a gluing criterion

In this section we record a practical criterion for proving that a functor on (Sch/S) is representable. The key point is that a functor which satisfies the sheaf condition for the Zariski topology and which is locally covered by representable open subfunctors can be obtained by gluing the corresponding schemes.

Throughout, S is a fixed base scheme, and all functors are contravariant functors

$$(\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}.$$

2.1. The sheaf condition for the Zariski topology.

We begin with the sheaf condition that will later allow us to glue local moduli data.

Definition 2.1. Let

$$F : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

be a contravariant functor. We say that F *satisfies the sheaf condition for the Zariski topology* if for every S-scheme T and every open covering $\{U_\alpha\}_{\alpha \in A}$ of T, the sequence

$$F(T) \longrightarrow \prod_{\alpha} F(U_\alpha) \rightrightarrows \prod_{\alpha,\beta} F(U_\alpha \cap U_\beta)$$

is an equaliser. Equivalently, the following two conditions hold:

(1) *Locality:* if $\xi, \eta \in F(T)$ have the same restriction to each U_α, then $\xi = \eta$;
(2) *Gluing:* if elements $\xi_\alpha \in F(U_\alpha)$ are given and satisfy

$$\xi_\alpha|_{U_\alpha \cap U_\beta} = \xi_\beta|_{U_\alpha \cap U_\beta}$$

for all α, β, then there exists a unique $\xi \in F(T)$ whose restriction to each U_α is ξ_α.

Lemma 2.2. *Every representable functor on* (Sch/S) *satisfies the sheaf condition for the Zariski topology.*

Proof. Suppose $F \cong h_X$ for some S-scheme X. Let T be an S-scheme and let $\{U_\alpha\}$ be an open covering of T.

An element of $h_X(T)$ is an S-morphism $f : T \to X$. If two such morphisms have the same restriction to each U_α, then they are equal, because morphisms of schemes are determined by their restrictions to an open cover. This proves locality.

Now suppose we are given morphisms $f_\alpha : U_\alpha \to X$ such that on each overlap $U_\alpha \cap U_\beta$ one has

$$f_\alpha|_{U_\alpha \cap U_\beta} = f_\beta|_{U_\alpha \cap U_\beta}.$$

By the usual gluing lemma for morphisms of schemes, there exists a unique morphism $f : T \to X$ restricting to all the f_α. This gives the required gluing statement. □

Remark 2.3. In moduli problems, this sheaf condition is usually a formal expression of the fact that families and isomorphisms of families can be glued Zariski locally. Later, for the Grassmannian functor, the relevant families will be quotient bundles or subbundles of a fixed vector bundle, and the sheaf condition will come from the local nature of vector bundles and short exact sequences.

2.2. **Subfunctors represented by open immersions.** We next isolate the notion of an open condition inside an arbitrary functor.

Definition 2.4. Let H and F be functors on (Sch/S). A *subfunctor* $H \subset F$ means a natural transformation $H \Rightarrow F$ such that for every S-scheme T the induced map

$$H(T) \longrightarrow F(T)$$

is injective. In this situation we usually identify $H(T)$ with a subset of $F(T)$.

Definition 2.5. Let $H \subset F$ be a subfunctor. We say that the inclusion $H \hookrightarrow F$ is *represented by open immersions* if for every S-scheme T and every element $\xi \in F(T)$ there exists an open subscheme $U_\xi \subset T$ with the following property:

for every morphism $f : T' \to T$ of S-schemes,

$$f \text{ factors through } U_\xi \qquad \Longleftrightarrow \qquad f^*\xi \in H(T').$$

Remark 2.6. Definition 2.5 is the functorial version of an open immersion. Indeed, if $F = h_X$ is representable and $\xi \in F(T) = \mathrm{Hom}_S(T, X)$ corresponds to a morphism $g : T \to X$, then the condition says precisely that the fiber product

$$h_T \times_F H$$

is represented by an open subscheme $U_\xi \subset T$. In other words, $H \to F$ behaves, after every base change from a scheme, like an open immersion of schemes.

Definition 2.7. A subfunctor $H \subset F$ is called an *open subfunctor* if the inclusion $H \hookrightarrow F$ is represented by open immersions.

Example 2.8. If $U \subset X$ is an open subscheme of an S-scheme X, then

$$h_U \subset h_X$$

is an open subfunctor.

Proof. Let T be an S-scheme and let $\xi \in h_X(T)$ correspond to a morphism $g : T \to X$. Define

$$U_\xi := g^{-1}(U) \subset T.$$

This is an open subscheme of T. For any $f : T' \to T$, the composite $g \circ f : T' \to X$ factors through U if and only if f factors through $g^{-1}(U) = U_\xi$. This is exactly the condition in Definition 2.5. □

Lemma 2.9. *Let $H_1, H_2 \subset F$ be open subfunctors. Then*

$$H_1 \cap H_2 \subset F$$

is also an open subfunctor.

Proof. Let T be an S-scheme and let $\xi \in F(T)$. Since each H_i is represented by open immersions, there are open subschemes $U_i \subset T$ such that for every $f : T' \to T$,

$$f^*\xi \in H_i(T') \quad \Longleftrightarrow \quad f \text{ factors through } U_i.$$

Then

$$f^*\xi \in (H_1 \cap H_2)(T')$$

if and only if f factors through both U_1 and U_2, hence if and only if f factors through $U_1 \cap U_2$. Therefore $H_1 \cap H_2$ is represented by the open immersion $U_1 \cap U_2 \hookrightarrow T$. □

2.3. **A representability criterion via an open cover.** We now formulate the criterion that will later be applied to the Grassmannian functor.

Definition 2.10. Let $\{F_i \subset F\}_{i \in I}$ be a family of subfunctors. We say that the family *covers* F if for every S-scheme T and every $\xi \in F(T)$ there exists an open covering

$$T = \bigcup_{\alpha \in A} U_\alpha$$

and indices $i(\alpha) \in I$ such that

$$\xi|_{U_\alpha} \in F_{i(\alpha)}(U_\alpha)$$

for every α.

Theorem 2.11 (Representability via an open cover)**.** *Let*

$$F : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

be a contravariant functor. Suppose that:

(1) F satisfies the sheaf condition for the Zariski topology;
(2) there exists a family of subfunctors $\{F_i \subset F\}_{i \in I}$ such that each F_i is representable;
(3) for each $i \in I$, the inclusion $F_i \hookrightarrow F$ is represented by open immersions;
(4) the family $\{F_i\}_{i \in I}$ covers F in the sense of Definition 2.10.

Then F is representable by an S-scheme.

2.4. **Proof of the gluing criterion.** We now give a complete proof of Theorem 2.11.

Proof of Theorem 2.11. For each $i \in I$, choose an S-scheme X_i representing F_i, and write

$$F_i \cong h_{X_i}.$$

Let

$$\xi_i \in F_i(X_i) \subset F(X_i)$$

be the element corresponding to the identity morphism $\mathrm{id}_{X_i} \in h_{X_i}(X_i)$.

Step 1: construction of the overlaps. Fix $i, j \in I$. Since the inclusion $F_j \hookrightarrow F$ is represented by open immersions, we may apply Definition 2.5 to the scheme X_i and the element $\xi_i \in F(X_i)$. We obtain an open subscheme

$$U_{ij} \subset X_i$$

such that for every morphism $f : T \to X_i$,

$$f \text{ factors through } U_{ij} \qquad \Longleftrightarrow \qquad f^*\xi_i \in F_j(T).$$

In particular, the inclusion $U_{ij} \hookrightarrow X_i$ shows that

$$\xi_i|_{U_{ij}} \in F_j(U_{ij}).$$

Since $F_j \cong h_{X_j}$, there is a unique morphism

$$\varphi_{ij} : U_{ij} \longrightarrow X_j$$

such that

$$\varphi_{ij}^*\xi_j = \xi_i|_{U_{ij}}.$$

We claim that φ_{ij} factors through $U_{ji} \subset X_j$. Indeed, $\xi_i|_{U_{ij}}$ lies not only in $F_j(U_{ij})$ by construction, but also in $F_i(U_{ij})$ because it is the restriction of $\xi_i \in F_i(X_i)$. Hence, by the defining property of U_{ji}, the morphism φ_{ij} factors through U_{ji}. We will still denote the induced morphism by

$$\varphi_{ij} : U_{ij} \longrightarrow U_{ji}.$$

Step 2: the transition maps are isomorphisms. We show that

$$\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$$

is an isomorphism with inverse φ_{ji}.

Consider the composite

$$U_{ij} \xrightarrow{\varphi_{ij}} U_{ji} \hookrightarrow X_j \xrightarrow{\varphi_{ji}} X_i.$$

Pulling back ξ_i along this composite gives

$$(\varphi_{ji} \circ \varphi_{ij})^*\xi_i = \varphi_{ij}^*(\varphi_{ji}^*\xi_i) = \varphi_{ij}^*(\xi_j|_{U_{ji}}) = \varphi_{ij}^*\xi_j = \xi_i|_{U_{ij}}.$$

But the identity map on U_{ij} also has this property. Since $F_i \cong h_{X_i}$, morphisms from U_{ij} to X_i are uniquely determined by the corresponding elements of $F_i(U_{ij})$. Therefore

$$\varphi_{ji} \circ \varphi_{ij} = \mathrm{id}_{U_{ij}}.$$

Similarly,

$$\varphi_{ij} \circ \varphi_{ji} = \mathrm{id}_{U_{ji}}.$$

Thus φ_{ij} is an isomorphism with inverse φ_{ji}.

Step 3: the cocycle condition. For $i, j, k \in I$, we claim that on the common domain of definition the transition maps satisfy

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}.$$

Indeed, both sides are morphisms from the open subset

$$U_{ij} \cap U_{ik} \subset X_i$$

to X_k. Pulling back ξ_k along either side gives

$$(\varphi_{jk} \circ \varphi_{ij})^*\xi_k = \varphi_{ij}^*(\varphi_{jk}^*\xi_k) = \varphi_{ij}^*(\xi_j|_{U_{jk}}) = \xi_i|_{U_{ij}\cap U_{ik}},$$

and also

$$\varphi_{ik}^*\xi_k = \xi_i|_{U_{ij}\cap U_{ik}}.$$

Since $F_k \cong h_{X_k}$, the two morphisms are equal.

Therefore the data $\{X_i, U_{ij}, \varphi_{ij}\}$ satisfy the usual compatibility conditions for gluing schemes along open subsets.

Step 4: gluing the schemes. By the gluing theorem for schemes, there exists an S-scheme X together with open immersions

$$\psi_i : X_i \hookrightarrow X$$

such that

$$\psi_i(X_i) \cap \psi_j(X_j) = \psi_i(U_{ij})$$

and the identifications on overlaps are given by φ_{ij}. Moreover, the open subschemes $\psi_i(X_i)$ cover X.

Step 5: gluing the local universal elements. We claim that the elements $\xi_i \in F(X_i)$ glue to a global element

$$\xi \in F(X).$$

On the overlap $\psi_i(U_{ij}) = \psi_j(U_{ji})$, the two restrictions correspond via φ_{ij}, because by construction

$$\varphi_{ij}^* \xi_j = \xi_i|_{U_{ij}}.$$

Since F satisfies the sheaf condition for the Zariski topology, the ξ_i glue uniquely to an element $\xi \in F(X)$.

This element defines a natural transformation

$$\eta : h_X \longrightarrow F, \qquad \eta_T(f) = f^*\xi.$$

Step 6: surjectivity of η_T. Let T be an S-scheme and let $\zeta \in F(T)$. Because the family $\{F_i\}$ covers F, there exists an open covering

$$T = \bigcup_\alpha T_\alpha$$

and indices $i(\alpha)$ such that

$$\zeta|_{T_\alpha} \in F_{i(\alpha)}(T_\alpha).$$

Since $F_{i(\alpha)} \cong h_{X_{i(\alpha)}}$, each restricted element determines a unique morphism

$$f_\alpha : T_\alpha \longrightarrow X_{i(\alpha)}$$

satisfying

$$f_\alpha^* \xi_{i(\alpha)} = \zeta|_{T_\alpha}.$$

We claim that the f_α agree on overlaps after passing to the glued scheme X. Let $T_{\alpha\beta} = T_\alpha \cap T_\beta$. On this overlap, both morphisms correspond to the same element

$$\zeta|_{T_{\alpha\beta}} \in F(T_{\alpha\beta}).$$

Since $f_\alpha^* \xi_{i(\alpha)} = \zeta|_{T_{\alpha\beta}}$ lies in $F_{i(\beta)}(T_{\alpha\beta})$, the defining property of the open subset

$$U_{i(\alpha)i(\beta)} \subset X_{i(\alpha)}$$

shows that $f_\alpha|_{T_{\alpha\beta}}$ factors through $U_{i(\alpha)i(\beta)}$. Similarly $f_\beta|_{T_{\alpha\beta}}$ factors through $U_{i(\beta)i(\alpha)}$. Under the identification

$$\varphi_{i(\alpha)i(\beta)} : U_{i(\alpha)i(\beta)} \xrightarrow{\sim} U_{i(\beta)i(\alpha)},$$

the two maps agree, because both correspond to the same element of $F_{i(\beta)}(T_{\alpha\beta})$. Hence the morphisms $\psi_{i(\alpha)} \circ f_\alpha : T_\alpha \to X$ glue to a unique morphism

$$f : T \longrightarrow X.$$

By construction,

$$f^*\xi = \zeta.$$

Thus η_T is surjective.

Step 7: injectivity of η_T. Now suppose $f, g : T \to X$ satisfy

$$f^*\xi = g^*\xi \in F(T).$$

We show that $f = g$.

Since the open subschemes $\psi_i(X_i)$ cover X, the opens

$$W_{ij} := f^{-1}(\psi_i(X_i)) \cap g^{-1}(\psi_j(X_j))$$

cover T. It suffices to show that f and g agree on each W_{ij}.

Fix i, j. On W_{ij}, the map f factors through X_i and the map g factors through X_j. Write

$$\widetilde{f} : W_{ij} \to X_i, \qquad \widetilde{g} : W_{ij} \to X_j$$

for the induced morphisms. Since

$$\widetilde{f}^*\xi_i = f^*\xi|_{W_{ij}} = g^*\xi|_{W_{ij}} = \widetilde{g}^*\xi_j,$$

the common element lies in $F_i(W_{ij})$. By the defining property of $U_{ji} \subset X_j$, the morphism $\widetilde{g}$ factors through U_{ji}. Similarly, $\widetilde{f}$ factors through U_{ij}.

Now both morphisms

$$\widetilde{f} : W_{ij} \to X_i \qquad \text{and} \qquad \varphi_{ji} \circ \widetilde{g} : W_{ij} \to X_i$$

pull back ξ_i to the same element of $F_i(W_{ij})$. Since $F_i \cong h_{X_i}$, these two morphisms are equal. Therefore $\widetilde{f}$ and $\widetilde{g}$ agree after passing to the glued scheme X. Hence

$$f|_{W_{ij}} = g|_{W_{ij}}.$$

Since the W_{ij} cover T, we conclude that $f = g$.

Thus $\eta_T : h_X(T) \to F(T)$ is injective.

We have shown that η_T is bijective for every T, and it is natural in T by construction. Therefore

$$h_X \xrightarrow{\sim} F,$$

so F is representable. □

Remark 2.12. The theorem says that representability can be proved in two stages. One first identifies local pieces of the moduli problem that are already represented by schemes, and then one uses the sheaf condition to glue these local pieces into a global representing object. This is exactly the strategy we will use for the Grassmannian.

3. The Grassmannian moduli problem

We now introduce the moduli problem represented by the Grassmannian. Throughout this section, S is a fixed base scheme, E is a finite locally free $\mathcal{O}_S$-module of rank n, and d is an integer with $0 < d < n$.

Our basic convention is that for an S-scheme $T \xrightarrow{g} S$ we write

$$E_T := g^*E.$$

Although one may describe the Grassmannian either by subbundles or by quotient bundles, we will take the quotient-bundle point of view as the primary one. This will make the local chart construction in the next section more transparent.

3.1. **Vector bundles, subbundles, quotient bundles, and exact sequences.** We begin by fixing the basic linear-algebraic language on schemes.

Definition 3.1. Let X be a scheme.

(1) A *vector bundle of rank* r on X is a finite locally free $\mathcal{O}_X$-module $\mathcal{E}$ of rank r.
(2) If $\mathcal{E}$ is a vector bundle on X, a *quotient bundle* of $\mathcal{E}$ of rank m is a surjective morphism
$$q : \mathcal{E} \twoheadrightarrow \mathcal{Q}$$
where $\mathcal{Q}$ is a vector bundle of rank m.
(3) A *subbundle* of $\mathcal{E}$ of rank d is a subsheaf
$$\mathcal{U} \subset \mathcal{E}$$
such that $\mathcal{U}$ is a vector bundle of rank d and the quotient $\mathcal{E}/\mathcal{U}$ is a vector bundle.

Remark 3.2. The extra requirement that $\mathcal{E}/\mathcal{U}$ be locally free is essential. An arbitrary locally free subsheaf of a vector bundle need not define a well-behaved family in moduli problems, while subbundles in the above sense are exactly the kernels of quotient bundles.

Lemma 3.3. *Let X be a scheme, let $\mathcal{E}$ be a vector bundle of rank n on X, and let*
$$q : \mathcal{E} \twoheadrightarrow \mathcal{Q}$$
be a quotient bundle of rank $n-d$. Then the kernel
$$\mathcal{U} := \ker(q)$$
is a subbundle of $\mathcal{E}$ of rank d, and there is a short exact sequence
$$0 \longrightarrow \mathcal{U} \longrightarrow \mathcal{E} \longrightarrow \mathcal{Q} \longrightarrow 0.$$

Proof. The claim is local on X, so we may work on an open set on which
$$\mathcal{E} \cong \mathcal{O}_X^{\oplus n} \qquad \text{and} \qquad \mathcal{Q} \cong \mathcal{O}_X^{\oplus (n-d)}.$$
Then q is given by a surjective matrix of functions
$$\mathcal{O}_X^{\oplus n} \longrightarrow \mathcal{O}_X^{\oplus (n-d)}.$$
At every point of X, surjectivity on stalks means that some $(n-d) \times (n-d)$ minor is invertible in a neighbourhood of that point. After restricting to such a neighbourhood, the matrix may be put into the form
$$\begin{bmatrix} I_{n-d} & A \end{bmatrix}$$
for some $(n-d) \times d$ matrix A. Its kernel is then visibly free of rank d, generated by the columns of
$$\begin{pmatrix} -A \\ I_d \end{pmatrix}.$$
Hence $\ker(q)$ is locally free of rank d, and the displayed sequence is exact. □

Lemma 3.4. *Let X be a scheme, let $\mathcal{E}$ be a vector bundle of rank n on X, and let $\mathcal{U} \subset \mathcal{E}$ be a subbundle of rank d. Then the quotient*
$$\mathcal{Q} := \mathcal{E}/\mathcal{U}$$
is a vector bundle of rank $n-d$, and there is a short exact sequence
$$0 \longrightarrow \mathcal{U} \longrightarrow \mathcal{E} \longrightarrow \mathcal{Q} \longrightarrow 0.$$

Proof. This is immediate from the definition of subbundle. □

Remark 3.5. Thus quotient bundles, subbundles, and short exact sequences with locally free terms are three equivalent ways to encode the same geometric data. We will define the Grassmannian moduli functor using quotient bundles and then pass to subbundles when this is more convenient.

3.2. **The Grassmannian functor.** We now define the functor that the Grassmannian is meant to represent.

Definition 3.6. The *Grassmannian functor* associated to (E, d) is the contravariant functor

$$\mathcal{G}_{d,E} : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

defined as follows.

For an S-scheme $T \xrightarrow{g} S$, the set $\mathcal{G}_{d,E}(T)$ consists of isomorphism classes of quotient bundles

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

where $\mathcal{Q}$ is a finite locally free $\mathcal{O}_T$-module of rank $n - d$.

Two quotients

$$q : E_T \twoheadrightarrow \mathcal{Q}, \qquad q' : E_T \twoheadrightarrow \mathcal{Q}'$$

are identified if there exists an isomorphism

$$\varphi : \mathcal{Q} \xrightarrow{\sim} \mathcal{Q}'$$

such that

$$\varphi \circ q = q'.$$

If $f : T' \to T$ is a morphism of S-schemes, the pullback map

$$f^* : \mathcal{G}_{d,E}(T) \longrightarrow \mathcal{G}_{d,E}(T')$$

sends the class of

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

to the class of the pulled back quotient

$$f^*q : E_{T'} = f^*E_T \twoheadrightarrow f^*\mathcal{Q}.$$

Remark 3.7. Since pullback preserves finite locally free sheaves and is right exact, the construction above is well-defined. The use of isomorphism classes is unavoidable, because a quotient bundle is only determined up to a unique isomorphism of its target bundle.

Remark 3.8. The functor $\mathcal{G}_{d,E}$ is intended to parametrize rank d subbundles of E_T, but the quotient description is technically more convenient. In particular, on standard affine charts one may normalize the quotient map and read off explicit coordinates from the images of the remaining basis vectors.

3.3. **Equivalent reformulation via rank-d subbundles.** We now relate the quotient description to the subbundle description.

Proposition 3.9. *For every S-scheme T, there are natural bijections between the following sets:*

(1) the set $\mathcal{G}_{d,E}(T)$ of isomorphism classes of quotient bundles

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

with $\mathcal{Q}$ locally free of rank $n-d$;

(2) the set of rank d subbundles

$$\mathcal{U} \subset E_T;$$

(3) the set of isomorphism classes of short exact sequences

$$0 \longrightarrow \mathcal{U} \longrightarrow E_T \longrightarrow \mathcal{Q} \longrightarrow 0$$

in which $\mathcal{U}$ and $\mathcal{Q}$ are locally free of ranks d and $n-d$ respectively.

These identifications are functorial in T.

Proof. Given a quotient bundle

$$q : E_T \twoheadrightarrow \mathcal{Q},$$

Lemma 3.3 produces a subbundle

$$\mathcal{U} = \ker(q) \subset E_T$$

and a short exact sequence

$$0 \longrightarrow \mathcal{U} \longrightarrow E_T \longrightarrow \mathcal{Q} \longrightarrow 0.$$

Conversely, given a subbundle $\mathcal{U} \subset E_T$, Lemma 3.4 produces a quotient bundle

$$E_T \twoheadrightarrow E_T/\mathcal{U}.$$

These constructions are inverse to each other. The equivalence with short exact sequences is simply a reformulation of the same data.

Functoriality in T follows from the functoriality of pullback for kernels, cokernels, and exact sequences of locally free sheaves. □

Remark 3.10. Later we will move freely between these three descriptions. The quotient viewpoint is best suited to the construction of standard affine charts, while the subbundle viewpoint is often better for geometric intuition.

3.4. **Base change and local nature on the base.** The Grassmannian functor behaves well under change of base.

Lemma 3.11. *Let $f : S' \to S$ be a morphism of schemes, and let*

$$E_{S'} := f^*E.$$

Then for every S'-scheme T there is a natural identification

$$\mathcal{G}_{d,E}(T) \cong \mathcal{G}_{d,E_{S'}}(T),$$

where on the left T is viewed as an S-scheme via $T \to S' \to S$. Equivalently, the restriction of $\mathcal{G}_{d,E}$ to (Sch/S') is canonically isomorphic to $\mathcal{G}_{d,E_{S'}}$.

Proof. If $T \to S'$ is an S'-scheme, then

$$(E_{S'})_T \cong E_T$$

canonically. Hence a quotient bundle

$$E_T \twoheadrightarrow \mathcal{Q}$$

of rank $n-d$ is exactly the same datum as a quotient bundle

$$(E_{S'})_T \twoheadrightarrow \mathcal{Q}$$

of rank $n-d$. The identification is clearly functorial in T. □

Remark 3.12. Lemma 3.11 shows that the moduli problem depends only on the pullback of E to the relevant base. In particular, if E becomes trivial on an open subset of S, then the Grassmannian functor over that open subset becomes the standard Grassmannian functor for a trivial bundle.

Lemma 3.13. *Let $\{U_\alpha\}_{\alpha\in A}$ be an open covering of S. Suppose that for each α, the functor*

$$\mathcal{G}_{d,E|_{U_\alpha}} : (\mathrm{Sch}/U_\alpha)^{\mathrm{op}} \to \mathbf{Set}$$

is representable by a U_α-scheme. Then representability of $\mathcal{G}_{d,E}$ is a Zariski-local question on S: the local representing schemes over the U_α glue on overlaps, and the resulting S-scheme represents $\mathcal{G}_{d,E}$.

Proof. By Lemma 3.11, the restriction of $\mathcal{G}_{d,E}$ to (Sch/U_α) is canonically identified with $\mathcal{G}_{d,E|_{U_\alpha}}$. On overlaps $U_\alpha \cap U_\beta$, the two resulting representing schemes represent the same functor, so by Corollary 1.7 they are canonically isomorphic. These canonical isomorphisms satisfy the cocycle condition, again by uniqueness. Hence the local representing schemes glue to an S-scheme representing $\mathcal{G}_{d,E}$. □

Remark 3.14. Since every finite locally free sheaf is locally trivial in the Zariski topology, Lemma 3.13 reduces the construction of $\mathrm{Gr}_S(d,E)$ to the case

$$E \cong \mathcal{O}_S^{\oplus n}.$$

This is the case in which the standard affine charts can be written down explicitly.

3.5. **Zariski descent for the Grassmannian functor.** We now show that the Grassmannian functor satisfies the sheaf condition for the Zariski topology.

Proposition 3.15. *The functor*

$$\mathcal{G}_{d,E} : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

satisfies the sheaf condition for the Zariski topology.

Proof. Let T be an S-scheme and let $\{U_\alpha\}_{\alpha\in A}$ be an open covering of T.

Locality. Suppose two elements of $\mathcal{G}_{d,E}(T)$ have the same restriction to each U_α. Choose quotient representatives

$$q : E_T \twoheadrightarrow \mathcal{Q}, \qquad q' : E_T \twoheadrightarrow \mathcal{Q}'.$$

The hypothesis means that for every α there exists an isomorphism

$$\varphi_\alpha : \mathcal{Q}|_{U_\alpha} \xrightarrow{\sim} \mathcal{Q}'|_{U_\alpha}$$

such that

$$\varphi_\alpha \circ q|_{U_\alpha} = q'|_{U_\alpha}.$$

Because $q|_{U_\alpha}$ is surjective, such an isomorphism is unique if it exists. Hence the φ_α agree on overlaps $U_\alpha \cap U_\beta$, and therefore glue to a unique global isomorphism

$$\varphi : \mathcal{Q} \xrightarrow{\sim} \mathcal{Q}'$$

satisfying

$$\varphi \circ q = q'.$$

Thus the two quotient bundles define the same element of $\mathcal{G}_{d,E}(T)$.

Gluing. Now suppose we are given elements

$$\xi_\alpha \in \mathcal{G}_{d,E}(U_\alpha)$$

whose restrictions to each overlap agree. Choose quotient representatives

$$q_\alpha : E_{U_\alpha} \twoheadrightarrow \mathcal{Q}_\alpha$$

with $\mathcal{Q}_\alpha$ locally free of rank $n-d$.

The agreement of the classes on overlaps means that for every pair α, β there exists an isomorphism

$$\varphi_{\alpha\beta} : \mathcal{Q}_\alpha|_{U_\alpha \cap U_\beta} \xrightarrow{\sim} \mathcal{Q}_\beta|_{U_\alpha \cap U_\beta}$$

such that

$$\varphi_{\alpha\beta} \circ q_\alpha|_{U_\alpha \cap U_\beta} = q_\beta|_{U_\alpha \cap U_\beta}.$$

Again, surjectivity implies that each $\varphi_{\alpha\beta}$ is unique. Hence on triple overlaps one automatically has

$$\varphi_{\beta\gamma} \circ \varphi_{\alpha\beta} = \varphi_{\alpha\gamma}.$$

Therefore the locally free sheaves $\mathcal{Q}_\alpha$ glue to a quasi-coherent sheaf $\mathcal{Q}$ on T, and the local morphisms q_α glue to a morphism

$$q : E_T \longrightarrow \mathcal{Q}.$$

Since surjectivity may be checked on stalks, q is surjective because each q_α is surjective. Since local freeness is also a local property, $\mathcal{Q}$ is a vector bundle of rank $n-d$. Thus

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

defines an element of $\mathcal{G}_{d,E}(T)$ whose restriction to each U_α is ξ_α.

Uniqueness follows from the locality statement already proved. □

Remark 3.16. Proposition 3.15 says that the Grassmannian moduli problem behaves like the functor of points of a scheme with respect to Zariski localization. Combined with the gluing criterion of Section 2, this is exactly the input needed to build the Grassmannian from explicit affine charts.

4. Standard open subfunctors and affine charts

In this section we construct the local charts for the Grassmannian functor. The main point is that after trivializing the ambient bundle, the moduli problem admits a natural open cover by subfunctors on which the universal quotient can be written in a normal form. These local pieces are represented by affine spaces, and their overlaps are governed by explicit transition maps.

4.1. **Reduction to the trivial bundle.** By Lemma 3.13, representability of the Grassmannian functor is a Zariski-local question on the base. Since every finite locally free sheaf is Zariski-locally trivial, it is enough to understand the case

$$E = \mathcal{O}_S^{\oplus n}.$$

Notation 4.1. For the remainder of this section we write

$$\mathcal{G}_{d,n} := \mathcal{G}_{d,\mathcal{O}_S^{\oplus n}}.$$

Thus for an S-scheme T, the set $\mathcal{G}_{d,n}(T)$ consists of isomorphism classes of quotient bundles

$$q : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{Q}$$

with $\mathcal{Q}$ locally free of rank $n-d$.

Remark 4.2. Although the Grassmannian may equally well be described by rank d subbundles of $\mathcal{O}_T^{\oplus n}$, the quotient-bundle description is especially convenient for constructing affine charts. The reason is that on a suitable open set one can choose a distinguished set of $n-d$ basis vectors whose images form a basis of the quotient bundle, and this yields a canonical normal form for the quotient map.

4.2. **Standard open subfunctors defined by invertible minors.** We now define the standard open pieces of $\mathcal{G}_{d,n}$.

Fix a subset

$$I = \{i_1 < \cdots < i_{n-d}\} \subset \{1, \ldots, n\}$$

of cardinality $n-d$, and let

$$J = \{j_1 < \cdots < j_d\}$$

be its complement. For any S-scheme T, write

$$\mathcal{O}_T^{\oplus n} = \mathcal{O}_T^I \oplus \mathcal{O}_T^J \cong \mathcal{O}_T^{\oplus(n-d)} \oplus \mathcal{O}_T^{\oplus d},$$

where $\mathcal{O}_T^I$ is the direct summand spanned by the standard basis vectors indexed by I, and similarly for $\mathcal{O}_T^J$.

Definition 4.3. The *standard open subfunctor* associated to I is the subfunctor

$$\mathcal{G}_I \subset \mathcal{G}_{d,n}$$

defined as follows. For an S-scheme T, an element of $\mathcal{G}_I(T)$ is an isomorphism class of quotient bundles

$$q : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{Q}$$

such that the induced morphism

$$q_I : \mathcal{O}_T^I \longrightarrow \mathcal{Q}$$

is an isomorphism of vector bundles.

Remark 4.4. Equivalently, if one chooses a local trivialization of $\mathcal{Q}$ and writes q as an $(n-d)\times n$ matrix, then the condition $q \in \mathcal{G}_I(T)$ means that the minor consisting of the columns indexed by I is invertible. Thus the standard open subfunctors are the functorial analogues of the usual Grassmannian charts defined by a nonvanishing maximal minor.

4.3. **These subfunctors are represented by open immersions.** We next show that each $\mathcal{G}_I$ defines an open condition on the Grassmannian functor in the precise sense of Section 2.

Proposition 4.5. *For each subset $I \subset \{1, \dots, n\}$ of cardinality $n-d$, the inclusion*

$$\mathcal{G}_I \hookrightarrow \mathcal{G}_{d,n}$$

is represented by open immersions.

Proof. Let T be an S-scheme and let

$$\xi = [q : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{Q}] \in \mathcal{G}_{d,n}(T).$$

Consider the induced morphism

$$q_I : \mathcal{O}_T^I \longrightarrow \mathcal{Q}.$$

Both source and target are vector bundles of rank $n-d$. Hence the set of points $t \in T$ at which $(q_I)_t$ is an isomorphism is an open subset of T, namely the locus where the determinant of q_I is invertible. Let

$$U_{I,\xi} \subset T$$

denote this open subset.

We claim that $U_{I,\xi}$ has the property required by Definition 2.5. Let $f : T' \to T$ be a morphism of S-schemes. Then the pullback of ξ is represented by

$$f^*q : \mathcal{O}_{T'}^{\oplus n} \twoheadrightarrow f^*\mathcal{Q},$$

and the induced morphism on the I-summand is

$$(f^*q)_I = f^*(q_I) : \mathcal{O}_{T'}^I \longrightarrow f^*\mathcal{Q}.$$

Since a morphism between vector bundles of the same rank is an isomorphism exactly on the inverse image of its isomorphism locus, we have

$$f^*\xi \in \mathcal{G}_I(T') \iff f^*(q_I) \text{ is an isomorphism} \iff f \text{ factors through } U_{I,\xi}.$$

This proves that $\mathcal{G}_I \hookrightarrow \mathcal{G}_{d,n}$ is represented by open immersions. □

Proposition 4.6. *The family of standard open subfunctors*

$$\{\mathcal{G}_I\}_{|I|=n-d}$$

covers $\mathcal{G}_{d,n}$.

Proof. Let

$$\xi = [q : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{Q}] \in \mathcal{G}_{d,n}(T).$$

Fix a point $t \in T$. Since $q_t : k(t)^n \twoheadrightarrow \mathcal{Q}(t)$ is a surjective linear map onto an $(n-d)$-dimensional vector space, some set of $n-d$ standard basis vectors of $k(t)^n$ maps to a basis of $\mathcal{Q}(t)$. Equivalently, there exists a subset $I \subset \{1, \dots, n\}$ of cardinality $n-d$ such that the fiber of

$$q_I : \mathcal{O}_T^I \longrightarrow \mathcal{Q}$$

at t is an isomorphism.

Because the isomorphism locus of q_I is open, there is an open neighbourhood $V_t \subset T$ of t such that

$$\xi|_{V_t} \in \mathcal{G}_I(V_t).$$

As t varies, these open neighbourhoods form an open covering of T, so the collection $\{\mathcal{G}_I\}$ covers $\mathcal{G}_{d,n}$ in the sense of Definition 2.10. □

4.4. **Explicit affine coordinates on each standard chart.** We now show that each standard open subfunctor is represented by an affine space of dimension $d(n-d)$ over S.

Fix I and J as above. Set

$$m = d(n-d),$$

and write

$$\mathbb{A}^m_S = \underline{\mathrm{Spec}}_S\big(\mathcal{O}_S[x_{\alpha\beta}]_{1\le\alpha\le n-d,\ 1\le\beta\le d}\big).$$

We think of the functions $x_{\alpha\beta}$ as the entries of an $(n-d)\times d$ matrix

$$X = (x_{\alpha\beta}).$$

Proposition 4.7. *For each subset $I \subset \{1,\dots,n\}$ of cardinality $n-d$, the standard open subfunctor $\mathcal{G}_I$ is representable by $\mathbb{A}^{d(n-d)}_S$.*

Proof. Let T be an S-scheme.

We first define a map

$$\Psi_T : \mathrm{Hom}_S(T, \mathbb{A}^m_S) \longrightarrow \mathcal{G}_I(T).$$

A morphism $\varphi : T \to \mathbb{A}^m_S$ is equivalent to a choice of functions

$$a_{\alpha\beta} \in \Gamma(T, \mathcal{O}_T), \qquad 1 \le \alpha \le n-d,\ 1 \le \beta \le d,$$

that is, to an $(n-d)\times d$ matrix

$$A = (a_{\alpha\beta}).$$

Using the decomposition

$$\mathcal{O}_T^{\oplus n} = \mathcal{O}_T^I \oplus \mathcal{O}_T^J,$$

define a morphism

$$q_A : \mathcal{O}_T^I \oplus \mathcal{O}_T^J \longrightarrow \mathcal{O}_T^I$$

by

$$q_A(u,v) = u + Av.$$

Since the restriction of q_A to $\mathcal{O}_T^I$ is the identity, q_A is surjective and defines an element of $\mathcal{G}_I(T)$:

$$\Psi_T(\varphi) := [q_A : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{O}_T^I].$$

Conversely, let

$$[q : \mathcal{O}_T^{\oplus n} \twoheadrightarrow \mathcal{Q}] \in \mathcal{G}_I(T).$$

By definition, the induced map

$$q_I : \mathcal{O}_T^I \xrightarrow{\sim} \mathcal{Q}$$

is an isomorphism. Using this isomorphism to identify $\mathcal{Q} \cong \mathcal{O}_T^I$, the quotient q becomes a surjective morphism

$$\widetilde{q} : \mathcal{O}_T^I \oplus \mathcal{O}_T^J \longrightarrow \mathcal{O}_T^I$$

whose restriction to the first summand is the identity. Therefore $\widetilde{q}$ is uniquely of the form

$$\widetilde{q}(u,v) = u + Av$$

for a uniquely determined matrix

$$A = (a_{\alpha\beta}) \in M_{(n-d)\times d}\big(\Gamma(T, \mathcal{O}_T)\big).$$

These functions define a morphism

$$\Phi_T([q]) : T \longrightarrow \mathbb{A}^m_S.$$

It is immediate that Φ_T and Ψ_T are inverse to one another. Moreover, the construction is functorial in T: pulling back the quotient q_A along a morphism $f : T' \to T$ replaces the matrix A by its pullback f^*A. Hence the bijections Φ_T assemble into a natural isomorphism

$$\mathcal{G}_I \xrightarrow{\sim} h_{\mathbb{A}^m_S}.$$

□

Remark 4.8. Under the identification of Proposition 4.7, a point of $\mathcal{G}_I(T)$ is encoded by the *off-pivot coefficients* of a quotient matrix in normal form

$$\begin{bmatrix} I_{n-d} & A \end{bmatrix}.$$

Thus each standard chart is an affine space of dimension

$$(n-d)d = d(n-d),$$

as expected.

4.5. **The universal quotient and universal subbundle on a chart.** The affine description above makes the universal family completely explicit.

Notation 4.9. Let

$$X_I := \mathbb{A}_S^{d(n-d)}$$

be the scheme representing $\mathcal{G}_I$, and let

$$X = (x_{\alpha\beta})$$

denote the universal $(n-d) \times d$ matrix of coordinate functions on X_I. We write

$$\mathcal{Q}_I := \mathcal{O}^I_{X_I}$$

for the universal quotient bundle on the chart X_I.

Proposition 4.10. *On the chart X_I there is a canonical short exact sequence*

$$0 \longrightarrow \mathcal{U}_I \longrightarrow \mathcal{O}^{\oplus n}_{X_I} \xrightarrow{q_I^{\mathrm{univ}}} \mathcal{Q}_I \longrightarrow 0$$

representing the universal element of $\mathcal{G}_I(X_I)$, where

$$q_I^{\mathrm{univ}}(u, v) = u + Xv$$

under the decomposition

$$\mathcal{O}^{\oplus n}_{X_I} = \mathcal{O}^I_{X_I} \oplus \mathcal{O}^J_{X_I}.$$

Moreover, the kernel $\mathcal{U}_I$ is a rank d subbundle of $\mathcal{O}^{\oplus n}_{X_I}$ given as the image of the injective map

$$\mathcal{O}^J_{X_I} \longrightarrow \mathcal{O}^I_{X_I} \oplus \mathcal{O}^J_{X_I}, \qquad v \longmapsto (-Xv, v).$$

Proof. By Proposition 4.7, the scheme X_I represents the functor $\mathcal{G}_I$. The identity morphism on X_I corresponds to the universal matrix X, hence to the quotient

$$q_I^{\mathrm{univ}} : \mathcal{O}^I_{X_I} \oplus \mathcal{O}^J_{X_I} \longrightarrow \mathcal{Q}_I = \mathcal{O}^I_{X_I}, \qquad (u, v) \longmapsto u + Xv.$$

This is surjective because its restriction to the first summand is the identity.

Its kernel consists exactly of pairs (u, v) satisfying $u = -Xv$, so it is the image of the map

$$v \longmapsto (-Xv, v).$$

This identifies the kernel with a trivial rank d vector bundle, hence produces the displayed short exact sequence. □

Remark 4.11. On each standard chart, the universal quotient bundle is trivial and the universal subbundle is the graph of the matrix $-X$. This is the local normal form that will later glue to the global universal exact sequence on the Grassmannian.

4.6. **Overlaps and transition functions.** We now explain how the affine charts meet and how the transition maps arise.

Fix two subsets

$$I, I' \subset \{1, \dots, n\}, \qquad |I| = |I'| = n - d,$$

and let J and J' be their complements. On the chart X_I, the universal quotient is given by a universal matrix

$$M_I(X)$$

of size $(n-d) \times n$ whose columns indexed by I form the identity matrix and whose columns indexed by J are the columns of the universal matrix X.

Let

$$M_{I,I'}(X)$$

denote the submatrix of $M_I(X)$ formed by the columns indexed by I'.

Proposition 4.12. *Under the identification $X_I \cong \mathcal{G}_I$, the intersection*

$$\mathcal{G}_I \cap \mathcal{G}_{I'}$$

is represented by the principal open subset

$$X_{I,I'} := D\big(\det(M_{I,I'}(X))\big) \subset X_I.$$

Proof. Let T be an S-scheme and let

$$f : T \to X_I$$

correspond to a quotient in $\mathcal{G}_I(T)$ represented in normal form by a matrix

$$M_I(A),$$

where $A = f^*X$ is an $(n-d) \times d$ matrix of functions on T. The pullback quotient lies in $\mathcal{G}_{I'}(T)$ if and only if the columns indexed by I' form a basis of the quotient bundle, that is, if and only if the induced map

$$\mathcal{O}_T^{I'} \longrightarrow \mathcal{Q}$$

is an isomorphism. In matrix terms this means exactly that the submatrix

$$M_{I,I'}(A) = f^* M_{I,I'}(X)$$

is invertible. Since invertibility of a square matrix of functions is equivalent to invertibility of its determinant, this holds if and only if f factors through the principal open subset

$$D\big(\det(M_{I,I'}(X))\big) \subset X_I.$$

Hence the intersection $\mathcal{G}_I \cap \mathcal{G}_{I'}$ is represented by $X_{I,I'}$. □

On this overlap one may re-express the same quotient in the normal form adapted to I'. This yields the transition map.

Proposition 4.13. *There is a canonical isomorphism*

$$\theta_{I,I'} : X_{I,I'} \xrightarrow{\sim} X_{I',I}$$

characterized by the fact that the pullback of the universal quotient on $X_{I'}$ along $\theta_{I,I'}$ is identified with the restriction of the universal quotient on X_I to $X_{I,I'}$.

More concretely, on $X_{I,I'}$ the universal quotient matrix $M_I(X)$ may be renormalized by left multiplication with the inverse of $M_{I,I'}(X)$:

$$M_{I'}(X') = M_{I,I'}(X)^{-1} M_I(X),$$

where $M_{I'}(X')$ is in the normal form adapted to I'. The entries of the new matrix X' are regular functions on $X_{I,I'}$, and they define the morphism $\theta_{I,I'}$.

Proof. On the principal open subset $X_{I,I'}$, the matrix $M_{I,I'}(X)$ is invertible. Therefore

$$M_{I,I'}(X)^{-1} M_I(X)$$

is a well-defined matrix of regular functions on $X_{I,I'}$. By construction, its columns indexed by I' form the identity matrix, so it is in the normal form corresponding to the chart I'. The remaining columns determine an $(n-d) \times d$ matrix X' of regular functions on $X_{I,I'}$, hence a morphism

$$\theta_{I,I'} : X_{I,I'} \longrightarrow X_{I'}.$$

This morphism lands in $X_{I',I}$ because the same quotient is also in the chart I. By symmetry, one constructs a morphism

$$\theta_{I',I} : X_{I',I} \longrightarrow X_I.$$

Both compositions correspond to leaving the quotient family unchanged and then rewriting it back in the original normal form. Therefore they are the identity by the uniqueness of the normal form on each chart. Hence $\theta_{I,I'}$ is an isomorphism with inverse $\theta_{I',I}$. □

Corollary 4.14. *The transition isomorphisms $\theta_{I,I'}$ satisfy the cocycle condition on triple overlaps.*

Proof. On a triple overlap, the three relevant charts all represent the same restriction of the functor $\mathcal{G}_{d,n}$. The transition maps are characterized by the requirement that they identify the corresponding universal quotient families. Hence the composite

$$\theta_{I',I''} \circ \theta_{I,I'}$$

and the direct map

$$\theta_{I,I''}$$

induce the same quotient family, so they are equal by the uniqueness of the representing morphism. □

Remark 4.15. The content of this section is that the Grassmannian functor for a trivial bundle is locally modelled by affine spaces with explicit matrix coordinates, and that the overlap maps are given by the usual change of pivot minor. In the next section we will apply the gluing criterion of Section 2 to these charts and thereby construct the Grassmannian $\mathrm{Gr}_S(d,n)$ and its universal exact sequence.

5. Representability of the Grassmannian and the universal exact sequence

We now pass from the local constructions of Section 4 to the global representability of the Grassmannian functor. We first treat the case of a trivial bundle, where the standard open subfunctors provide an explicit affine atlas, and then we globalize to an arbitrary vector bundle by Zariski descent.

5.1. Construction of $\mathrm{Gr}_S(d,n)$ by gluing affine charts.

We begin with the Grassmannian functor

$$\mathcal{G}_{d,n} : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

for the trivial rank n bundle.

Theorem 5.1. *The functor $\mathcal{G}_{d,n}$ is representable by an S-scheme, denoted*

$$\mathrm{Gr}_S(d,n).$$

Moreover, for each subset

$$I \subset \{1,\dots,n\}, \qquad |I| = n-d,$$

the standard open subfunctor $\mathcal{G}_I \subset \mathcal{G}_{d,n}$ corresponds to an open subscheme

$$\mathrm{Gr}_S(d,n)_I \subset \mathrm{Gr}_S(d,n),$$

and each $\mathrm{Gr}_S(d,n)_I$ is canonically isomorphic to the affine space $\mathbb{A}_S^{d(n-d)}$.

Proof. By Proposition 3.15, the functor $\mathcal{G}_{d,n}$ satisfies the sheaf condition for the Zariski topology. By Proposition 4.7, each standard open subfunctor $\mathcal{G}_I$ is representable by $\mathbb{A}_S^{d(n-d)}$. By Proposition 4.5, the inclusion

$$\mathcal{G}_I \hookrightarrow \mathcal{G}_{d,n}$$

is represented by open immersions, and by Proposition 4.6 the family $\{\mathcal{G}_I\}$ covers $\mathcal{G}_{d,n}$.

Therefore all hypotheses of the representability criterion (Theorem 2.11) are satisfied. It follows that $\mathcal{G}_{d,n}$ is representable by an S-scheme, which we denote by $\mathrm{Gr}_S(d,n)$.

The final statement is built into the gluing construction: under the natural isomorphism

$$h_{\mathrm{Gr}_S(d,n)} \xrightarrow{\sim} \mathcal{G}_{d,n},$$

the subfunctor $\mathcal{G}_I$ is represented by an open subscheme $\mathrm{Gr}_S(d,n)_I \subset \mathrm{Gr}_S(d,n)$, and $\mathrm{Gr}_S(d,n)_I \cong \mathbb{A}_S^{d(n-d)}$ by Proposition 4.7. □

Remark 5.2. The gluing criterion abstracts exactly what happens in the classical matrix construction of the Grassmannian. The standard charts $\mathrm{Gr}_S(d,n)_I$ are affine spaces whose coordinates record the off-pivot entries of a quotient matrix in normal form, and the overlap maps are given by changing the pivot minor.

5.2. **Representability for a general vector bundle E.** We now return to an arbitrary finite locally free $\mathcal{O}_S$-module E of rank n.

Theorem 5.3. *Let E be a finite locally free $\mathcal{O}_S$-module of rank n, and let $0 < d < n$. Then the Grassmannian functor*

$$\mathcal{G}_{d,E} : (\mathrm{Sch}/S)^{\mathrm{op}} \longrightarrow \mathbf{Set}$$

is representable by an S-scheme, denoted

$$\mathrm{Gr}_S(d, E).$$

Proof. Choose an open covering $\{U_\alpha\}_{\alpha \in A}$ of S such that

$$E|_{U_\alpha} \cong \mathcal{O}_{U_\alpha}^{\oplus n}$$

for every α. By Lemma 3.11, the restriction of $\mathcal{G}_{d,E}$ to (Sch/U_α) is canonically identified with

$$\mathcal{G}_{d,E|_{U_\alpha}} \cong \mathcal{G}_{d,n}$$

over U_α. By Theorem 5.1, this functor is representable by a U_α-scheme, namely

$$\mathrm{Gr}_{U_\alpha}(d, n).$$

On overlaps $U_{\alpha\beta} := U_\alpha \cap U_\beta$, the restrictions of these local representing schemes represent the same functor

$$\mathcal{G}_{d,E|_{U_{\alpha\beta}}}$$

and hence are canonically isomorphic by Corollary 1.7. These canonical isomorphisms satisfy the cocycle condition, again by uniqueness. Therefore the local schemes $\mathrm{Gr}_{U_\alpha}(d, n)$ glue to an S-scheme

$$\mathrm{Gr}_S(d, E)$$

representing $\mathcal{G}_{d,E}$. □

Remark 5.4. Theorem 5.3 shows that the Grassmannian of a vector bundle is locally the same as the ordinary Grassmannian of a trivial bundle. In particular, all local properties that can be checked on the standard affine charts reduce to the trivial case.

5.3. **The universal quotient sequence.** Once representability is known, the universal family appears automatically by evaluating the moduli functor on the representing scheme itself.

Notation 5.5. Let

$$\pi : \mathrm{Gr}_S(d, E) \longrightarrow S$$

be the structure morphism.

Proposition 5.6. *There exists a canonical quotient bundle*

$$q_{\mathrm{univ}} : \pi^* E \twoheadrightarrow \mathcal{Q}_{\mathrm{univ}}$$

of rank $n - d$ on $\mathrm{Gr}_S(d, E)$ corresponding to the identity morphism

$$\mathrm{id}_{\mathrm{Gr}_S(d,E)} \in \mathrm{Hom}_S\big(\mathrm{Gr}_S(d, E), \mathrm{Gr}_S(d, E)\big)$$

under the natural bijection

$$\mathrm{Hom}_S\big(T, \mathrm{Gr}_S(d, E)\big) \;\cong\; \mathcal{G}_{d,E}(T).$$

If we define

$$\mathcal{U}_{\mathrm{univ}} := \ker(q_{\mathrm{univ}}),$$

then there is a short exact sequence

$$0 \longrightarrow \mathcal{U}_{\mathrm{univ}} \longrightarrow \pi^* E \xrightarrow{q_{\mathrm{univ}}} \mathcal{Q}_{\mathrm{univ}} \longrightarrow 0 \tag{5.7}$$

in which $\mathcal{U}_{\mathrm{univ}}$ *and* $\mathcal{Q}_{\mathrm{univ}}$ *are vector bundles of ranks* d *and* $n-d$.

Proof. Since $\mathrm{Gr}_S(d,E)$ represents $\mathcal{G}_{d,E}$, the identity morphism corresponds to an element of

$$\mathcal{G}_{d,E}\big(\mathrm{Gr}_S(d,E)\big),$$

that is, to an isomorphism class of quotient bundles

$$q_{\mathrm{univ}} : \pi^* E \twoheadrightarrow \mathcal{Q}_{\mathrm{univ}}$$

of rank $n-d$.

By Lemma 3.3, its kernel

$$\mathcal{U}_{\mathrm{univ}} := \ker(q_{\mathrm{univ}})$$

is a vector bundle of rank d, and we obtain the short exact sequence (5.7). □

Remark 5.8. On the standard chart $\mathrm{Gr}_S(d,n)_I$, the universal quotient sequence restricts to the explicit sequence constructed in Proposition 4.10. Thus the global universal quotient is obtained by gluing the local normal forms from the affine charts.

5.4. **The fine moduli interpretation.** The universal quotient sequence gives the expected classifying property of the Grassmannian.

Proposition 5.9. *Let* $T \xrightarrow{g} S$ *be an* S*-scheme. Then the assignment*

$$f \longmapsto f^*(q_{\mathrm{univ}})$$

induces a bijection between

$$\mathrm{Hom}_S\big(T, \mathrm{Gr}_S(d,E)\big)$$

and the set of isomorphism classes of quotient bundles

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

with $\mathcal{Q}$ *locally free of rank* $n-d$.

Equivalently, every such quotient bundle on T *arises uniquely as the pullback of the universal quotient along a unique morphism*

$$f : T \longrightarrow \mathrm{Gr}_S(d,E).$$

Proof. This is simply the statement that $\mathrm{Gr}_S(d,E)$ represents the functor $\mathcal{G}_{d,E}$. Under the representing isomorphism

$$h_{\mathrm{Gr}_S(d,E)} \xrightarrow{\sim} \mathcal{G}_{d,E},$$

a morphism

$$f : T \to \mathrm{Gr}_S(d,E)$$

corresponds to the pullback of the universal element

$$[q_{\mathrm{univ}} : \pi^* E \twoheadrightarrow \mathcal{Q}_{\mathrm{univ}}] \in \mathcal{G}_{d,E}\big(\mathrm{Gr}_S(d,E)\big)$$

along f. □

Remark 5.10. In this sense $\mathrm{Gr}_S(d,E)$ is a *fine moduli space* for rank $n-d$ quotient bundles of E, or equivalently for rank d subbundles of E with locally free quotient.

5.5. **Base change for Grassmann bundles.** The construction of Grassmann bundles is compatible with pullback on the base.

Proposition 5.11. *Let $f : S' \to S$ be a morphism of schemes, and let*

$$E_{S'} := f^*E.$$

Then there is a canonical isomorphism of S'-schemes

$$\mathrm{Gr}_{S'}(d, E_{S'}) \xrightarrow{\sim} \mathrm{Gr}_S(d, E) \times_S S'.$$

Under this identification, the universal quotient on $\mathrm{Gr}_{S'}(d, E_{S'})$ is the pullback of the universal quotient on $\mathrm{Gr}_S(d, E)$.

Proof. By Lemma 3.11, the restriction of the functor $\mathcal{G}_{d,E}$ to (Sch/S') is canonically isomorphic to $\mathcal{G}_{d,E_{S'}}$. On the other hand, the fiber product

$$\mathrm{Gr}_S(d, E) \times_S S'$$

represents the restriction of $\mathcal{G}_{d,E}$ to (Sch/S'), because its functor of points over S' is given by

$$T \longmapsto \mathrm{Hom}_{S'}\big(T, \mathrm{Gr}_S(d, E) \times_S S'\big) \cong \mathrm{Hom}_S\big(T, \mathrm{Gr}_S(d, E)\big).$$

Hence both $\mathrm{Gr}_{S'}(d, E_{S'})$ and $\mathrm{Gr}_S(d, E) \times_S S'$ represent the same functor on (Sch/S'), so they are canonically isomorphic by Corollary 1.7.

The compatibility of universal quotients is obtained by tracing the identity element through this identification. Equivalently, both quotients represent the same universal element of the same functor. □

Remark 5.12. Proposition 5.11 says that forming the Grassmann bundle commutes with base change. This is one of the main reasons that Grassmann bundles behave so well in families.

5.6. **Functoriality under bundle isomorphisms.** Finally, the construction depends functorially on the vector bundle itself, at least with respect to isomorphisms.

Proposition 5.13. *Let E and E' be finite locally free $\mathcal{O}_S$-modules of rank n, and let*

$$\varphi : E \xrightarrow{\sim} E'$$

be an isomorphism. Then there is a canonical isomorphism of S-schemes

$$\mathrm{Gr}_S(d, E) \xrightarrow{\sim} \mathrm{Gr}_S(d, E').$$

It is characterized by the property that a quotient bundle

$$q : E_T \twoheadrightarrow \mathcal{Q}$$

corresponds to the quotient bundle

$$q \circ \varphi_T^{-1} : E'_T \twoheadrightarrow \mathcal{Q}$$

for every S-scheme T.

Proof. For every S-scheme T, the pullback of φ gives an isomorphism

$$\varphi_T : E_T \xrightarrow{\sim} E'_T.$$

Hence transport of structure along φ_T defines a bijection between quotient bundles of E_T of rank $n - d$ and quotient bundles of E'_T of rank $n - d$:

$$[q : E_T \twoheadrightarrow \mathcal{Q}] \longmapsto [q \circ \varphi_T^{-1} : E'_T \twoheadrightarrow \mathcal{Q}].$$

These bijections are functorial in T, so they define a natural isomorphism of functors

$$\mathcal{G}_{d,E} \xrightarrow{\sim} \mathcal{G}_{d,E'}.$$

Since both functors are representable, the desired isomorphism of representing schemes follows from Corollary 1.7. □

Remark 5.14. In particular, a choice of local trivialization of E identifies the restriction of $\mathrm{Gr}_S(d,E)$ with the standard Grassmannian $\mathrm{Gr}_U(d,n)$ on that open set. This is the geometric content of the local triviality argument used in the proof of Theorem 5.3.

6. Tangent spaces and the relative tangent bundle

In this section we explain how the moduli interpretation of the Grassmannian controls its infinitesimal geometry. At a k-point, a tangent vector is a first-order deformation of a d-plane inside $V = k^n$, and the space of such deformations is naturally $\mathrm{Hom}_k(U, V/U)$. Globally, the same description produces the relative tangent bundle

$$T_{\mathrm{Gr}_S(d,E)/S} \cong \mathcal{H}om(\mathcal{U}_{\mathrm{univ}}, \mathcal{Q}_{\mathrm{univ}}).$$

6.1. **Tangent vectors as first-order deformations.** We begin with the usual description of tangent vectors by means of dual numbers.

Definition 6.1. Let k be a field. The *dual numbers over* k are the k-algebra

$$k[\varepsilon]/(\varepsilon^2).$$

We write

$$D_k := \mathrm{Spec}\big(k[\varepsilon]/(\varepsilon^2)\big).$$

Definition 6.2. Let X be a k-scheme and let $x \in X(k)$ be a k-point. The *Zariski tangent space* of X at x is the set

$$T_x X := \{k\text{-morphisms } \widetilde{x} : D_k \to X \text{ whose restriction to the closed point is } x\}.$$

Remark 6.3. The set $T_x X$ carries a natural structure of k-vector space. For the purposes of this section, the important point is the interpretation of its elements as first-order deformations of the point x over the dual numbers.

Remark 6.4. If X is a moduli space, then a morphism $D_k \to X$ is best viewed as a family over D_k whose special fiber is the object represented by the point x. In the case of the Grassmannian, this means that tangent vectors are first-order deformations of a quotient bundle, or equivalently of a subspace, over the dual numbers.

6.2. **Tangent space at a k-point.** We now specialize to the classical Grassmannian over a field.

Theorem 6.5. *Let k be a field, let V be an n-dimensional k-vector space, and let*

$$G = \mathrm{Gr}_k(d, V).$$

Let $[U] \in G(k)$ be the point corresponding to a d-dimensional subspace $U \subset V$. Then there is a natural isomorphism of k-vector spaces

$$T_{[U]}G \xrightarrow{\sim} \mathrm{Hom}_k(U, V/U).$$

Proof. A tangent vector at $[U]$ is, by definition, a morphism

$$\widetilde{x} : D_k \to G$$

whose special fiber is the point $[U]$. Using the subbundle interpretation of the Grassmannian, this is equivalent to giving a rank d subbundle

$$\widetilde{U} \subset V \otimes_k k[\varepsilon]$$

whose reduction modulo ε is $U \subset V$.

Choose a basis $u_1, \ldots, u_d$ of U. Since $\widetilde{U}$ is free of rank d over the local ring $k[\varepsilon]/(\varepsilon^2)$, after choosing lifts of the basis vectors we may write $\widetilde{U}$ as the submodule generated by

$$u_i + \varepsilon v_i, \qquad v_i \in V, \qquad i = 1, \ldots, d.$$

The classes of the v_i in V/U define a k-linear map

$$\phi_{\widetilde{U}} : U \longrightarrow V/U, \qquad u_i \longmapsto v_i \bmod U.$$

We claim that this map depends only on the deformation $\widetilde{U}$. Indeed, replacing v_i by another lift $v_i + u_i'$ with $u_i' \in U$ changes $u_i + \varepsilon v_i$ by

$$u_i + \varepsilon(v_i + u_i') = (u_i + \varepsilon u_i') + \varepsilon v_i,$$

and this corresponds only to changing the chosen basis of the free $k[\varepsilon]$-module $\widetilde{U}$. Hence the resulting class in V/U is unchanged.

Conversely, let

$$\phi : U \longrightarrow V/U$$

be a k-linear map. Choose a basis $u_1, \ldots, u_d$ of U and choose vectors $v_i \in V$ lifting $\phi(u_i) \in V/U$. Let $\widetilde{U}_\phi$ be the $k[\varepsilon]$-submodule of $V \otimes_k k[\varepsilon]$ generated by

$$u_i + \varepsilon v_i, \qquad i = 1, \ldots, d.$$

This is a free rank d submodule whose reduction modulo ε is U. If we choose different lifts v_i, the generators are changed by an invertible $k[\varepsilon]$-linear change of basis, so the resulting submodule depends only on ϕ.

These constructions are inverse to one another, and they are clearly functorial in U. Therefore

$$T_{[U]}G \cong \operatorname{Hom}_k(U, V/U).$$

□

Corollary 6.6. *With the notation of Theorem 6.5, one has*

$$\dim_k T_{[U]}G = d(n-d).$$

Proof. Since $\dim_k U = d$ and $\dim_k(V/U) = n - d$, we have

$$\dim_k \operatorname{Hom}_k(U, V/U) = d(n-d).$$

The conclusion now follows from Theorem 6.5. □

Remark 6.7. The theorem says that an infinitesimal deformation of U is obtained by tilting U inside V in the normal directions V/U. This is exactly what one expects geometrically, and it is the fiberwise version of the global description of the relative tangent bundle below.

6.3. **The relative tangent bundle.** We now globalize the preceding tangent-space calculation.

Notation 6.8. We write

$$\pi : \mathrm{Gr}_S(d,E) \longrightarrow S$$

for the Grassmann bundle, and

$$0 \longrightarrow \mathcal{U}_{\mathrm{univ}} \longrightarrow \pi^* E \longrightarrow \mathcal{Q}_{\mathrm{univ}} \longrightarrow 0$$

for the universal exact sequence constructed in Proposition 5.6.

We also write

$$T_{\mathrm{Gr}_S(d,E)/S} := \mathcal{H}om_{\mathcal{O}_{\mathrm{Gr}_S(d,E)}}\big(\Omega_{\mathrm{Gr}_S(d,E)/S}, \mathcal{O}_{\mathrm{Gr}_S(d,E)}\big)$$

for the relative tangent sheaf.

Theorem 6.9. *There is a canonical isomorphism of vector bundles on* $\mathrm{Gr}_S(d,E)$*:*

$$T_{\mathrm{Gr}_S(d,E)/S} \;\cong\; \mathcal{H}om(\mathcal{U}_{\mathrm{univ}}, \mathcal{Q}_{\mathrm{univ}}) \;\cong\; \mathcal{U}_{\mathrm{univ}}^{\vee} \otimes \mathcal{Q}_{\mathrm{univ}}.$$

Proof. The assertion is local on $\mathrm{Gr}_S(d,E)$, so by local triviality of E and by the standard affine atlas of Section 4, it is enough to prove the statement on a standard chart

$$X_I \cong \mathbb{A}_S^{d(n-d)}.$$

On such a chart, by Proposition 4.10, the universal quotient sequence has the form

$$0 \longrightarrow \mathcal{U}_I \longrightarrow \mathcal{O}_{X_I}^I \oplus \mathcal{O}_{X_I}^J \xrightarrow{q_I^{\mathrm{univ}}} \mathcal{Q}_I \longrightarrow 0,$$

where

$$q_I^{\mathrm{univ}}(u,v) = u + Xv$$

and $X = (x_{\alpha\beta})$ is the universal $(n-d)\times d$ matrix of coordinates on X_I.

By construction, the map

$$\mathcal{O}_{X_I}^J \longrightarrow \mathcal{U}_I, \qquad v \longmapsto (-Xv, v)$$

is an isomorphism. Hence

$$\mathcal{U}_I \cong \mathcal{O}_{X_I}^{\oplus d}, \qquad \mathcal{Q}_I \cong \mathcal{O}_{X_I}^I \cong \mathcal{O}_{X_I}^{\oplus(n-d)},$$

and therefore

$$\mathcal{H}om(\mathcal{U}_I, \mathcal{Q}_I) \cong \mathcal{H}om\big(\mathcal{O}_{X_I}^{\oplus d}, \mathcal{O}_{X_I}^{\oplus(n-d)}\big) \cong \mathcal{O}_{X_I}^{\oplus d(n-d)}.$$

On the other hand,

$$X_I \cong \mathbb{A}_S^{d(n-d)}$$

with coordinates $x_{\alpha\beta}$, so its relative tangent bundle is the free sheaf

$$T_{X_I/S} \cong \mathcal{O}_{X_I}^{\oplus d(n-d)}$$

with basis given by the derivations

$$\frac{\partial}{\partial x_{\alpha\beta}}.$$

We now identify these two free sheaves. Let

$$E_{\alpha\beta} \in \mathcal{H}om(\mathcal{U}_I, \mathcal{Q}_I)$$

be the matrix unit sending the β-th basis vector of $\mathcal{U}_I \cong \mathcal{O}_{X_I}^{\oplus d}$ to the α-th basis vector of $\mathcal{Q}_I \cong \mathcal{O}_{X_I}^{\oplus(n-d)}$, and sending all other basis vectors to zero. We define a morphism of vector bundles

$$\Phi_I : \mathcal{H}om(\mathcal{U}_I, \mathcal{Q}_I) \longrightarrow T_{X_I/S}$$

by

$$E_{\alpha\beta} \longmapsto \frac{\partial}{\partial x_{\alpha\beta}}.$$

Since both sides are free of the same rank, Φ_I is an isomorphism.

It remains to check that these local isomorphisms agree on overlaps. Let $X_{I,I'} \subset X_I$ be an overlap as in Proposition 4.12, and let

$$\theta_{I,I'} : X_{I,I'} \xrightarrow{\sim} X_{I',I}$$

be the transition isomorphism of Proposition 4.13. Fix a T-valued point of $X_{I,I'}$, corresponding to a quotient in normal form

$$q_A : \mathcal{O}_T^I \oplus \mathcal{O}_T^J \longrightarrow \mathcal{O}_T^I, \qquad q_A(u,v) = u + Av,$$

where A is an $(n-d) \times d$ matrix of functions on T.

A tangent vector at this point of X_I is represented by a first-order deformation

$$A \longmapsto A + \varepsilon B,$$

where B is again an $(n-d) \times d$ matrix. This gives a first-order deformation of the quotient

$$q_{A+\varepsilon B}(u,v) = u + (A + \varepsilon B)v.$$

Passing to kernels, the corresponding first-order deformation of the subbundle determines a map

$$\mathcal{U}_A \longrightarrow \mathcal{Q}_A$$

which, under the identifications

$$\mathcal{U}_A \cong \mathcal{O}_T^{\oplus d}, \qquad \mathcal{Q}_A \cong \mathcal{O}_T^{\oplus(n-d)},$$

is represented exactly by the matrix B. Indeed, if

$$\mathcal{U}_A = \{(-Av, v) \mid v \in \mathcal{O}_T^{\oplus d}\},$$

then replacing A by $A + \varepsilon B$ changes the graph by the first-order term Bv in the quotient direction.

Now on the overlap $X_{I,I'}$, the same quotient family is rewritten in the I'-normal form by left multiplication with the invertible matrix $M_{I,I'}(A)^{-1}$. Therefore the first-order variation B is changed by the same change of basis that identifies the local trivializations of the universal subbundle and universal quotient on the two charts. In particular, the matrix describing the tangent vector and the matrix describing the induced homomorphism

$$\mathcal{U}_{\mathrm{univ}} \longrightarrow \mathcal{Q}_{\mathrm{univ}}$$

transform by the same rule on overlaps.

Hence the local isomorphisms

$$\Phi_I : \mathcal{H}om(\mathcal{U}_I, \mathcal{Q}_I) \xrightarrow{\sim} T_{X_I/S}$$

are compatible with the transition maps and glue to a global isomorphism

$$T_{\mathrm{Gr}_S(d,E)/S} \cong \mathcal{H}om(\mathcal{U}_{\mathrm{univ}}, \mathcal{Q}_{\mathrm{univ}}).$$

Finally, since $\mathcal{U}_{\mathrm{univ}}$ is locally free, the usual tensor-Hom identification gives

$$\mathcal{H}om(\mathcal{U}_{\mathrm{univ}}, \mathcal{Q}_{\mathrm{univ}}) \cong \mathcal{U}_{\mathrm{univ}}^{\vee} \otimes \mathcal{Q}_{\mathrm{univ}}.$$

This completes the proof. □

Remark 6.10. Theorem 6.9 is the global form of Theorem 6.5. Fiberwise, the sheaf

$$\mathcal{H}om(\mathcal{U}_{\mathrm{univ}}, \mathcal{Q}_{\mathrm{univ}})$$

specializes to

$$\mathrm{Hom}_k(U, V/U)$$

at the point corresponding to a d-plane $U \subset V$.

6.4. **Smoothness and relative dimension.** We now record the main geometric consequence of the affine atlas and the relative tangent bundle calculation.

Corollary 6.11. *The structure morphism*

$$\pi : \mathrm{Gr}_S(d, E) \longrightarrow S$$

is smooth of relative dimension $d(n-d)$.

Proof. Smoothness is local on the source and the target. By local triviality of E on S and by Theorem 5.1, the scheme $\mathrm{Gr}_S(d, E)$ is covered by open subsets isomorphic to

$$\mathbb{A}_S^{d(n-d)}.$$

Since the projection

$$\mathbb{A}_S^{d(n-d)} \longrightarrow S$$

is smooth of relative dimension $d(n-d)$, it follows that

$$\pi : \mathrm{Gr}_S(d, E) \longrightarrow S$$

is smooth of relative dimension $d(n-d)$.

The same dimension is also reflected in the relative tangent bundle: Theorem 6.9 identifies $T_{\mathrm{Gr}_S(d,E)/S}$ with a vector bundle of rank

$$\mathrm{rk}(\mathcal{U}_{\mathrm{univ}}^{\vee} \otimes \mathcal{Q}_{\mathrm{univ}}) = d(n-d).$$

□

Corollary 6.12. *For every point* $s \in S$ *with residue field* $\kappa(s)$*, the fiber*

$$\mathrm{Gr}_S(d, E)_s$$

is a smooth $\kappa(s)$*-scheme of dimension* $d(n-d)$.

Proof. This is the fiberwise form of Corollary 6.11. □

Remark 6.13. The moduli description of the Grassmannian does more than construct the scheme. It also explains its infinitesimal geometry: first-order deformations are controlled by maps from the universal subbundle to the universal quotient bundle. This is the key structural fact behind many later constructions on Grassmannians.

7. The line bundle $\mathcal{O}_{\mathrm{Gr}}(1)$, the Plücker morphism, and projectivity

In this section we study the projective geometry naturally attached to the universal quotient sequence. The determinant of the universal quotient bundle defines a distinguished line bundle on the Grassmannian, and the corresponding quotient of an exterior power of E produces the relative Plücker morphism into a projective bundle. Over a field this recovers the classical Plücker embedding.

7.1. Determinants of the universal bundles.

Let

$$\pi : \mathrm{Gr}_S(d, E) \longrightarrow S$$

be the Grassmann bundle, and let

$$0 \longrightarrow \mathcal{U}_{\mathrm{univ}} \longrightarrow \pi^* E \longrightarrow \mathcal{Q}_{\mathrm{univ}} \longrightarrow 0$$

be the universal exact sequence.

Definition 7.1. The determinant line bundles associated to the universal exact sequence are

$$\det(\mathcal{U}_{\mathrm{univ}}) := \bigwedge^{d} \mathcal{U}_{\mathrm{univ}}, \qquad \det(\mathcal{Q}_{\mathrm{univ}}) := \bigwedge^{n-d} \mathcal{Q}_{\mathrm{univ}}.$$

Proposition 7.2. *There is a canonical isomorphism of line bundles*

$$\det(\mathcal{U}_{\mathrm{univ}}) \otimes \det(\mathcal{Q}_{\mathrm{univ}}) \;\cong\; \pi^* \det(E).$$

Equivalently,

$$\det(\mathcal{Q}_{\mathrm{univ}}) \;\cong\; \det(\mathcal{U}_{\mathrm{univ}})^\vee \otimes \pi^* \det(E).$$

Proof. Taking determinants in the universal exact sequence gives

$$\det(\pi^* E) \;\cong\; \det(\mathcal{U}_{\mathrm{univ}}) \otimes \det(\mathcal{Q}_{\mathrm{univ}}).$$

Since determinant commutes with pullback for vector bundles, we have

$$\det(\pi^* E) \cong \pi^* \det(E).$$

Combining these two isomorphisms yields the result. □

Remark 7.3. When E is trivial of rank n, the relation simplifies to

$$\det(\mathcal{Q}_{\mathrm{univ}}) \cong \det(\mathcal{U}_{\mathrm{univ}})^\vee.$$

Thus in the trivial case one may think of the distinguished line bundle either as the determinant of the universal quotient or as the dual of the determinant of the universal subbundle.

7.2. The tautological ample line bundle on the Grassmannian.

The determinant of the universal quotient will serve as the basic projective line bundle on the Grassmannian.

Definition 7.4. We define

$$\mathcal{O}_{\mathrm{Gr}}(1) := \det(\mathcal{Q}_{\mathrm{univ}}).$$

Remark 7.5. This is the line bundle naturally singled out by the quotient construction of the Grassmannian. Many authors refer to $\mathcal{O}_{\mathrm{Gr}}(1)$ as the *Plücker line bundle*.

Remark 7.6. The notation $\mathcal{O}_{\mathrm{Gr}}(1)$ is meant to parallel the tautological quotient line bundle on projective space. In the next subsection we will make this precise by exhibiting $\mathcal{O}_{\mathrm{Gr}}(1)$ as the pullback of $\mathcal{O}(1)$ from a projective bundle.

7.3. **The relative Plücker morphism.** We now construct the Plücker morphism in families.

Definition 7.7. Let $\mathcal{F}$ be a finite locally free $\mathcal{O}_S$-module. We write

$$\mathbf{P}_S(\mathcal{F}) := \underline{\mathrm{Proj}}_S(\mathrm{Sym}(\mathcal{F}))$$

for the associated projective bundle over S, with the convention that $\mathbf{P}_S(\mathcal{F})$ parametrizes invertible quotients of $\mathcal{F}$.

Remark 7.8. Thus, if $g : T \to S$ is an S-scheme, a morphism

$$T \longrightarrow \mathbf{P}_S(\mathcal{F})$$

corresponds to an invertible quotient of $g^*\mathcal{F}$ on T. We denote the tautological quotient line bundle on $\mathbf{P}_S(\mathcal{F})$ by

$$\mathcal{O}_{\mathbf{P}_S(\mathcal{F})}(1).$$

Taking the top exterior power of the universal quotient yields a line bundle quotient

$$\bigwedge^{n-d} \pi^* E \twoheadrightarrow \bigwedge^{n-d} \mathcal{Q}_{\mathrm{univ}} = \det(\mathcal{Q}_{\mathrm{univ}}) = \mathcal{O}_{\mathrm{Gr}}(1).$$

Since exterior powers commute with pullback, this may be rewritten as

$$\pi^* \left(\bigwedge^{n-d} E \right) \twoheadrightarrow \mathcal{O}_{\mathrm{Gr}}(1).$$

Definition 7.9. The *relative Plücker morphism* is the morphism

$$\iota_E : \mathrm{Gr}_S(d, E) \longrightarrow \mathbf{P}_S \left(\bigwedge^{n-d} E \right)$$

corresponding to the quotient

$$\pi^* \left(\bigwedge^{n-d} E \right) \twoheadrightarrow \mathcal{O}_{\mathrm{Gr}}(1).$$

Proposition 7.10. *The relative Plücker morphism satisfies*

$$\iota_E^* \mathcal{O}_{\mathbf{P}_S(\wedge^{n-d} E)}(1) \cong \mathcal{O}_{\mathrm{Gr}}(1).$$

Proof. This is the defining property of the projective bundle in the quotient convention: the morphism ι_E is classified by the quotient

$$\pi^* \left(\bigwedge^{n-d} E \right) \twoheadrightarrow \mathcal{O}_{\mathrm{Gr}}(1),$$

so the pullback of the tautological quotient line bundle is exactly $\mathcal{O}_{\mathrm{Gr}}(1)$. □

Theorem 7.11. *The relative Plücker morphism*

$$\iota_E : \mathrm{Gr}_S(d, E) \longrightarrow \mathbf{P}_S \left(\bigwedge^{n-d} E \right)$$

is a closed immersion.

Proof. The assertion is local on S, so we may assume that

$$E \cong \mathcal{O}_S^{\oplus n}.$$

Write

$$G := \mathrm{Gr}_S(d, n).$$

Choose the standard basis $e_1, \ldots, e_n$ of $\mathcal{O}_S^{\oplus n}$. Then the exterior products

$$e_I := e_{i_1} \wedge \cdots \wedge e_{i_{n-d}}$$

indexed by subsets $I \subset \{1, \ldots, n\}$ of cardinality $n - d$ form a basis of $\bigwedge^{n-d} \mathcal{O}_S^{\oplus n}$, and hence determine homogeneous coordinates

$$p_I$$

on the projective bundle

$$\mathbf{P}_S\left(\bigwedge^{n-d} \mathcal{O}_S^{\oplus n}\right).$$

Fix such a subset I, and let

$$G_I \cong X_I \cong \mathbb{A}_S^{d(n-d)}$$

be the corresponding standard affine chart of Section 4. On X_I, the universal quotient is represented by the universal matrix

$$M_I(X),$$

whose columns indexed by I form the identity matrix and whose remaining columns are the entries of the universal $(n-d) \times d$ matrix $X = (x_{\alpha\beta})$.

The Plücker morphism on X_I is described by the maximal minors of $M_I(X)$. Since the minor indexed by I is equal to 1, the image of X_I lies in the standard affine open

$$D_+(p_I) \subset \mathbf{P}_S\left(\bigwedge^{n-d} \mathcal{O}_S^{\oplus n}\right).$$

On this affine open, the coordinate ring is generated by the ratios

$$\frac{p_J}{p_I}, \qquad J \neq I.$$

Under the Plücker map, these ratios are sent to the maximal minors of $M_I(X)$ divided by the minor indexed by I, hence simply to the maximal minors of $M_I(X)$.

Now if J differs from I by replacing a single index of I by a single index of its complement, then the corresponding maximal minor of $M_I(X)$ is, up to sign, one of the coordinates $x_{\alpha\beta}$. Hence every coordinate $x_{\alpha\beta}$ lies in the image of the induced homomorphism from the coordinate ring of $D_+(p_I)$ to the coordinate ring of X_I. Therefore that homomorphism is surjective, and the restriction

$$X_I \longrightarrow D_+(p_I)$$

of the relative Plücker morphism is a closed immersion.

Since the open subsets X_I cover G, the morphism ι_E is a closed immersion globally. □

Corollary 7.12. *The structure morphism*

$$\pi : \mathrm{Gr}_S(d, E) \longrightarrow S$$

is projective, and the line bundle $\mathcal{O}_{\mathrm{Gr}}(1)$ is relatively very ample over S.

Proof. By Theorem 7.11, the Grassmannian is a closed subscheme of the projective bundle

$$\mathbf{P}_S\left(\bigwedge^{n-d} E\right),$$

which is projective over S. Hence π is projective. The pullback statement of Proposition 7.10 shows that $\mathcal{O}_{\mathrm{Gr}}(1)$ is the restriction of the tautological quotient line bundle on this projective bundle, hence is relatively very ample. □

7.4. **Over a field: classical Plücker embedding.** We now specialize to the case where the base is a field.

Let k be a field, let V be an n-dimensional k-vector space, and write

$$G = \mathrm{Gr}_k(d, V).$$

Using the quotient-bundle description, the relative Plücker morphism takes the form

$$\iota : G \longrightarrow \mathbf{P}_k\left(\bigwedge^{n-d} V\right).$$

Remark 7.13. With the quotient convention for projective space, this is the natural target. If one prefers the classical line convention, then after the usual dual identifications one recovers the familiar Plücker map into a projective space built from $\bigwedge^d V$.

Theorem 7.14. *The morphism*

$$\iota : \mathrm{Gr}_k(d, V) \longrightarrow \mathbf{P}_k\left(\bigwedge^{n-d} V\right)$$

is a closed immersion.

Proof. This is the special case of Theorem 7.11 when $S = \mathrm{Spec}(k)$. □

Corollary 7.15. *Over a field k, the Grassmannian $\mathrm{Gr}_k(d, V)$ is a projective variety, and the line bundle $\mathcal{O}_{\mathrm{Gr}}(1)$ is very ample.*

Proof. By Theorem 7.14, the Grassmannian is a closed subscheme of projective space. The pullback statement of Proposition 7.10 shows that $\mathcal{O}_{\mathrm{Gr}}(1)$ is the restriction of the hyperplane line bundle, hence is very ample. □

7.5. **Example: $\mathrm{Gr}(2, 4) \subset \mathbf{P}^5$.** We now spell out the Plücker coordinates in the smallest interesting example.

Let k be a field and let

$$G = \mathrm{Gr}_k(2, 4).$$

Since

$$\dim_k\left(\bigwedge^2 k^4\right) = 6,$$

the Plücker embedding identifies G with a closed subvariety of

$$\mathbf{P}^5_k = \mathbf{P}_k\left(\bigwedge^2 k^4\right).$$

Remark 7.16. For this example, we use the classical Plücker coordinates associated to the line spanned by $u_1 \wedge u_2 \in \bigwedge^2 k^4$. This is compatible with our earlier quotient-based description after choosing the standard basis of $\bigwedge^2 k^4$ and the corresponding homogeneous coordinates on the ambient projective space.

Choose the standard basis e_1, e_2, e_3, e_4 of k^4. The vectors

$$e_1 \wedge e_2,\ e_1 \wedge e_3,\ e_1 \wedge e_4,\ e_2 \wedge e_3,\ e_2 \wedge e_4,\ e_3 \wedge e_4$$

form a basis of $\bigwedge^2 k^4$, and we write the corresponding homogeneous coordinates on $\mathbf{P}^5$ as

$$[p_{12} : p_{13} : p_{14} : p_{23} : p_{24} : p_{34}].$$

On the standard affine chart $U_{12} \subset G$, a 2-plane may be written as the span of

$$u_1 = e_1 + ae_3 + be_4, \qquad u_2 = e_2 + ce_3 + de_4,$$

where $a, b, c, d \in k$. Then

$$u_1 \wedge u_2 = e_1 \wedge e_2 + c\, e_1 \wedge e_3 + d\, e_1 \wedge e_4 - a\, e_2 \wedge e_3 - b\, e_2 \wedge e_4 + (ad - bc)\, e_3 \wedge e_4.$$

Hence the Plücker coordinates on U_{12} are

$$[p_{12} : p_{13} : p_{14} : p_{23} : p_{24} : p_{34}] = [1 : c : d : -a : -b : ad - bc].$$

In particular, on the affine chart $p_{12} \neq 0$ of $\mathbf{P}^5$, the Grassmannian is parametrized by the coordinates

$$a = -\frac{p_{23}}{p_{12}}, \qquad b = -\frac{p_{24}}{p_{12}}, \qquad c = \frac{p_{13}}{p_{12}}, \qquad d = \frac{p_{14}}{p_{12}}.$$

Substituting the formula for p_{34} gives the relation

$$\frac{p_{34}}{p_{12}} = \frac{p_{14}p_{23} - p_{13}p_{24}}{p_{12}^2}.$$

After homogenizing, one obtains the single quadratic equation

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

Proposition 7.17. *The image of the Plücker embedding*

$$\mathrm{Gr}_k(2,4) \hookrightarrow \mathbf{P}^5_k$$

is the smooth quadric hypersurface

$$p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0.$$

Proof. The computation above shows that every point of the Grassmannian satisfies the displayed quadratic equation. Conversely, on each affine chart $p_{ij} \neq 0$, the same normalization argument as above shows that the equation cuts out exactly one copy of $\mathbb{A}^4$ with the expected transition maps. Hence the quadric hypersurface obtained from the equation is covered by the same affine charts as $\mathrm{Gr}_k(2,4)$, and the two coincide. □

Remark 7.18. This example is a good model for the general situation. The standard Grassmannian charts are affine spaces, the Plücker coordinates are the maximal minors of a matrix in normal form, and the defining equations of the image are the quadratic Plücker relations among these minors.

7.6. **Final remarks.** We conclude with two standard consequences that lie slightly beyond the main line of this article.

Remark 7.19. Over an algebraically closed field, one can go further and show that the Picard group of the Grassmannian is infinite cyclic, generated by the class of $\mathcal{O}_{\mathrm{Gr}}(1)$.

Remark 7.20. From the moduli-theoretic point of view, the role of $\mathcal{O}_{\mathrm{Gr}}(1)$ is especially natural: it is obtained by taking the determinant of the universal quotient bundle, and the relative Plücker morphism is induced by the universal determinant quotient

$$\bigwedge^{n-d} \pi^* E \twoheadrightarrow \mathcal{O}_{\mathrm{Gr}}(1).$$